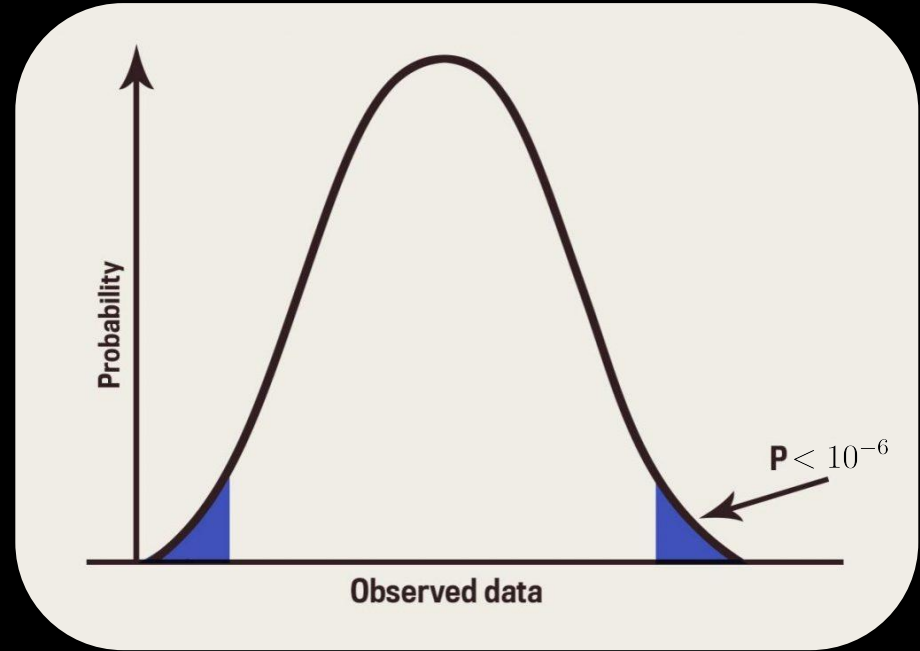


Quantum Sampler for Extremely Rare-Event Probability Estimation

Alyona Kosobokova
Brian J McDermott

Supported by WISER

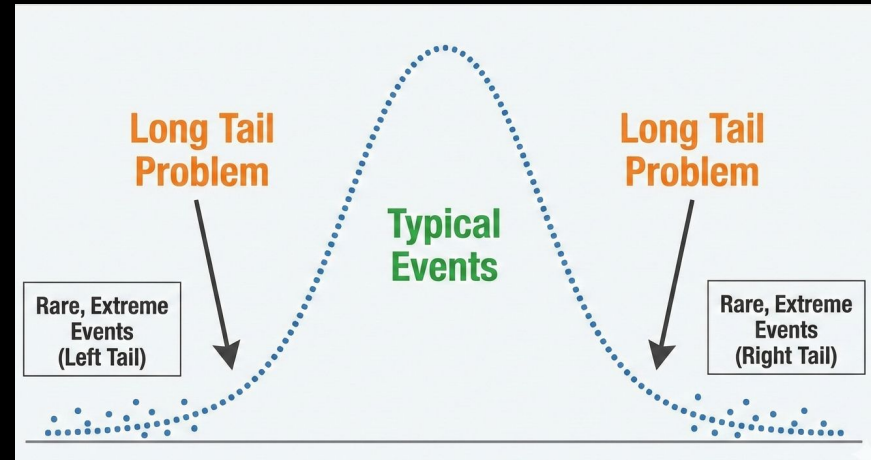


Problem Background

Ultra rare events with $p < 10^{-6}$, far tail of the distribution, many σ away from the mean.

Why do they matter?

- equipment failure (fault-tree analysis)
- cosmic rays, comets estimation
- extreme weather condition
- risk and safety modeling



Example: NASA requirements for the loss of crew probabilities are $p < 10^{-6}$ per mission.

Classical Approaches

Naive Monte Carlo becomes inefficient for $p \in [10^{-12}, 10^{-6}]$

Rare-event methods

- Importance sampling:
bias the sampling distribution and reweight samples.
- Splitting / Subset Simulation:
simulate toward the rare region using intermediate thresholds.
- Complex Fault and Event-Tree Analysis:
models with many failure pathways.



Requires problem-specific design.



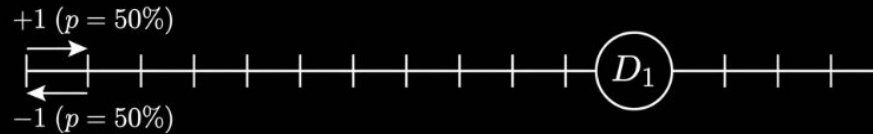
Research Question

Can we leverage Quantum Algorithms to improve the classical approach?

Problem Representation

1-D

- First-passage (hitting) probability: random walk with absorbing boundary.

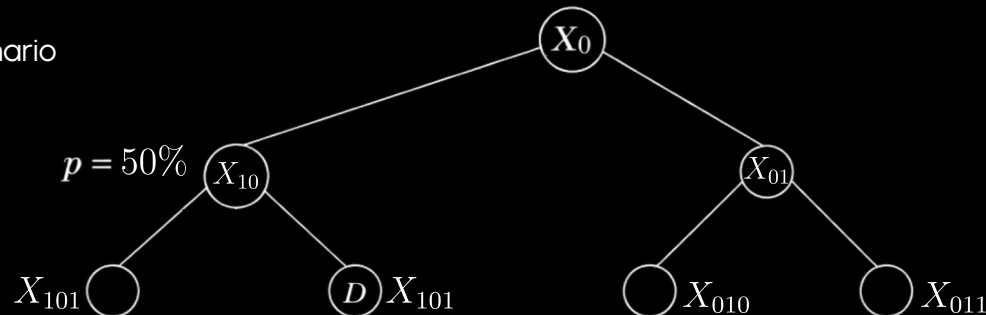


T - number of steps,
 D - absorbing boundary (depth).

When T is close to D , we observe extremely small hitting probability.

Multidimensional

- Fault tree scenario



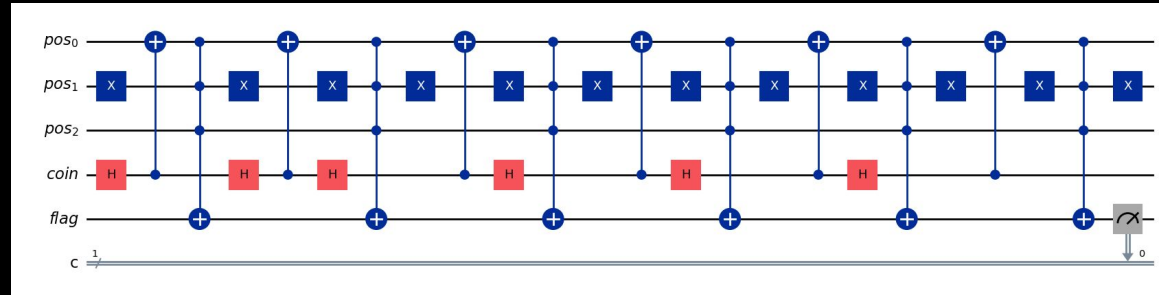
Quantum Methods

Problem Representation

$$x \in 0, 1, 2, 3, 4, 5 \quad \begin{array}{|c|c|c|c|c|c|} \hline & & & & & D_1 \\ \hline \end{array} X_0$$

Position	Binary
0	$ 000\rangle$
1	$ 001\rangle$
2	$ 010\rangle$
3	$ 011\rangle$
4	$ 100\rangle$
Flag:	5 $ 101\rangle$

Circuit Design



- Position register encodes walker location
- Coin qubit controls ± 1 step direction.
- Absorbing boundary marked with a flag qubit

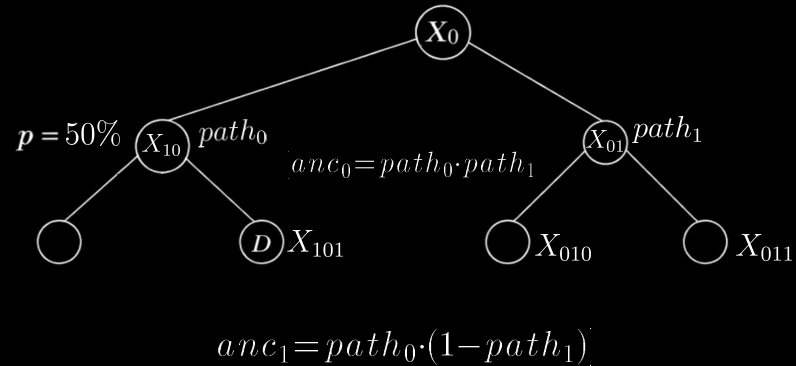


Does not scale well.

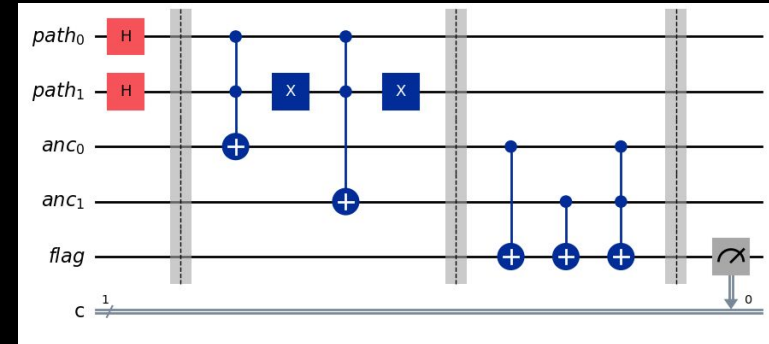


Quantum Methods

Problem Representation



Circuit Design



This circuit architecture also does not scale well.

$$N_{\text{qubits}} \approx 2D$$

Limitation and Challenges

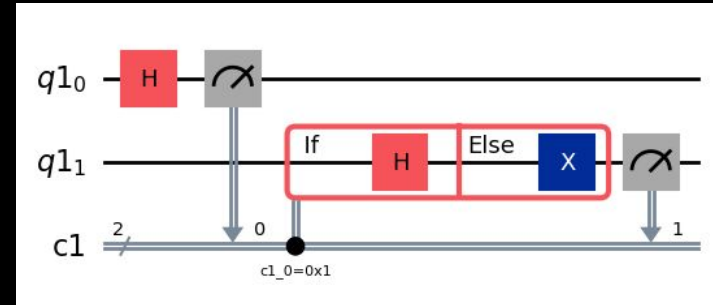
As the depth of the fault tree grows, the required number of gates, quantum registers, and error correction increases, restricting experiments to simplified toy models.

Follow up Work And Potential Solutions:

- Dynamic circuits to reuse quantum registers.
- Complex circuit design to reduce gate count and ancilla qubits.
- Combining classical and quantum algorithms,

Circuit optimization depends on the structure of the problem, such as a binary tree, graph, or one-dimensional walk. This conflicts with the objective of developing generic methods that do not require problem-specific tuning.

Dynamic circuits



Future Work Proposal

During this project, it became clear that there is a lack of short, modular educational material to help with constructing and testing quantum circuits in practice.

Proposal: Create short (under 10 minutes) videos with screencast and animation that cover from beginning to advance quantum computing concepts.

This can help practitioners move from theoretical descriptions of algorithms to working quantum circuits.

Topic Examples

- Quantum Gates Visually
- Quantum Circuits Types
- Oracle in QC
- Phase Kickback
- Amplitude Amplification
- Dynamic Circuits
- Ancilla Qubits as Memory
- Discrete vs Continuous QRW
- Coined vs Szegedy QRW
- Resource Scaling in OC



Thank you!

